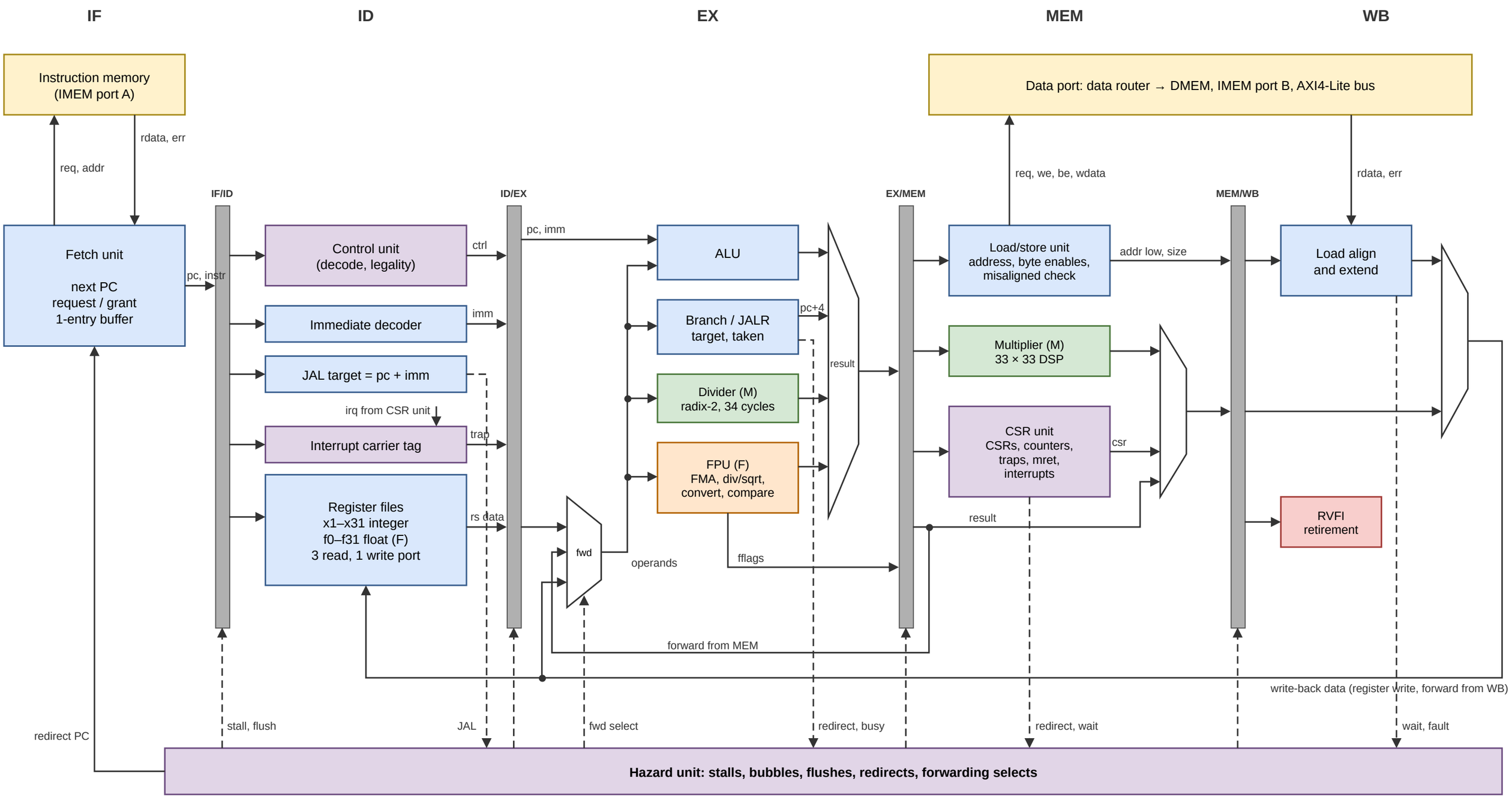
IF
ID
EX
MEM
WB
Instruction memory
(IMEM port A)
req, addr
rdata, err
Fetch unit
next PC
request / grant
1-entry buffer
pc, instr
IF/ID
Control unit
(decode, legality)
ctrl
Immediate decoder
imm
JAL target = pc + imm
irq from CSR unit
Interrupt carrier tag
trap
Register files
x1–x31 integer
f0–f31 float (F)
3 read, 1 write port
rs data
ID/EX
pc, imm
ALU
Branch / JALR
target, taken
pc+4
Divider (M)
radix-2, 34 cycles
FPU (F)
FMA, div/sqrt,
convert, compare
result
fwd
operands
fflags
EX/MEM
Data port: data router → DMEM, IMEM port B, AXI4-Lite bus
req, we, be, wdata
rdata, err
Load/store unit
address, byte enables,
misaligned check
addr low, size
Multiplier (M)
33 × 33 DSP
CSR unit
CSRs, counters,
traps, mret,
interrupts
csr
result
MEM/WB
Load align
and extend
RVFI
retirement
forward from MEM
write-back data (register write, forward from WB)
redirect PC
stall, flush
JAL
fwd select
redirect, busy
redirect, wait
wait, fault
Hazard unit: stalls, bubbles, flushes, redirects, forwarding selects